

ActInf GuestStream 093.1 ~ Nassim Dehouche "Enhancing Population-based Search with Active Inference"

GuestStream | Nassim Dehouche | 50:51

Hello and welcome. It's December 13th, 2024. We're in Active Inference Guest Stream 93.1 with Nassim Dehoush. We're going to be discussing a paper we wrote together, Enhancing Population-Based Search with Active Inference. So thank you, Nassim, for joining. Looking forward to your presentation and the discussion.

My pleasure. Thank you, Daniel. Very happy to be here and to share this joint work we did together a few months ago.

Perhaps I could give a brief background information about how this work came to be.

We did a podcast together and we had a few chats on Discord about Active Inference.

And it was actually a way you used to explain Active Inference.

You said that it's like the sense of touch, for example, wouldn't be really active unless you move your hand to really discover the texture.

And it's the idea that indeed perception is not a passive process where you're just receiving data and processing it.

It's that agents have an intent.

And this is something that meditators can also experience.

It's that even seeing, sight.

If you are really empty inside, wouldn't you really be active unless you have some intent.

So I thought about this in the context of a class of multi-agent systems that are used for optimization problems.

So perhaps I'll give an overview of our work.

So this is a kind of meta-heuristic methods that are used to solve all sorts of optimization problems.

But we mainly focus on combinatorial optimization, so problems in graphs typically.

And they use a population of agents to explore different solution paths.

And one common feature, as I will highlight later, of these methods is that they, just like the naive perception we have of how even humans perceive,

is that they consider that these agents have only reactive responses to the data they discover in the environment.

We'll see it in more details later when I do a brief literature review.

And so we try to incorporate predictive modeling, giving these agents an intent, which is to reduce their surprise and also update their belief.

We'll talk about that as well.

So the idea is to minimize the difference between the outcomes they observe and what they expect.

And we try to integrate this to this class of methods known as population-based search methods with an instantiation on one method called the ant colony optimization.

That's also inspired by you, Daniel, and your background working with ants.

And so this would be applied to a problem which I think is famous enough,

so I don't have to give a very detailed explanation of it,
but perhaps for people who are not familiar with the traveling salesman problem.
I will briefly explain it with this little app we put together for the paper.
So it's a problem in graphs where you have typically a set of cities or locations.
The problem consists in finding a route that visits all cities and goes back to your initial point.
Okay.

Now, with this kind of problems, there are two classes typically.
Problems in P, which can be solved in polynomial time.

One example would be finding the shortest route from one point to the other.
That is very easy to solve, and it remains easy even for very large-scale graphs,
very large numbers of cities.

But this problem, the traveling salesman problem,
the idea of visiting all cities and going back to your initial point,
happens to be in a different class, NP-hard.

So it is as hard as any computational problem you can think of.
And here again, the difficulty comes with scale.

It's that we don't have...

Pretty much we don't have a better method than exhaustively listing all parts.
So there are approximate solution methods known as heuristics and meta-heuristics.

A very simple one, a very intuitive one,
could be to visit the nearest city from your starting point
and repeat this process until you go back to your initial point.

So this is known as the nearest neighbor heuristic.

So here I would start, for example, from this one,
go to the next one, next one, next one, next one.

And then with this, you can get stuck in very bad branches of the tree of possible solutions.
For example, if the last city is very far here,
you would need to waste a lot of distance in your solution.

So nearest neighbor is not that great.

And the diameters that try to improve on it.

One of them is the one we will discuss in this presentation,
the ant-colonial optimization method.

So as I was saying, it's part of a large class of methods
known as population-based search meta-heuristics.

They are typically nature-inspired or biology-inspired.

But one very good point you raised, Daniel,
is that, well, can there be anything that is not nature-inspired?
That's a good question to ponder.

In any case, these are inspired by phenomena in nature.

Initially, I think the first method in this class

was invented in the 70s.

It's the genetic algorithm that is inspired by evolution.

And here again, there will be interesting bridges with active inference.

Ant-colony optimization is also one of the most successful and oldest.

But then there was a deluge of nature-inspired methods

for about 30 years.

Anything from honeybees to cuckoo to wolves to lions,

dolphins, dolphins, what have you,

even the way jazz players, human jazz players,

you know, synchronize when they improvise,

were used as inspiration for these methods.

And just as a fun fact,

artificial neural networks were considered

as part of this class of methods

up until the mid-2000s,

when I was an undergrad myself.

And they were usually listed in, you know,

textbooks like,

the two I have cited here,

among these population-based meta-historics

before taking, you know, life on their own.

But because of this, you know,

very large number of methods

that offer very little innovation,

some authors, notably Kenneth Sorensen here

with this important paper

that was published in 2013,

called for, you know,

a stop to these very superficial,

you know, improvements in your innovations,

just taking inspiration

from the next, you know, natural process.

And to go back to more abstract ways

of seeing these methods.

So more abstractly,

we can break down

these population-based search methods

in, you know,

by the fact that they have
these three characteristics.

First, there is, of course,
a population of agents
that cooperate.

So each one of them
is going to explore
some solution paths
since exploring, you know,
exhaustively exploring
the whole tree of possible solutions
is not feasible.

Agents would have some way
to explore solutions themselves
and they're going to cooperate,
share information.

Another common characteristics
of these methods
is that they usually
include self-adaptation.

The agents respond
to their environment
and we'll come back
to this point later
and modify their, you know,
their search strategy accordingly.

And there is also competition
among the solutions
discovered so far
by the population of agents
and we keep the best,
you know, subset of solutions.

So if we see it this way,
many of the, you know,
nature-inspired methods
we have here

come down to just ways
to, you know,
define these three characteristics, right?
But one common thing
among all methods pretty much
is that they all assume,
and I think it's a common bias
we have in designing algorithms
and multi-agent systems
is that we assume
that agents passively
receive information
and in the way
they retrieve this information,
they have no intent, right?
And so that's one thing
that we could possibly,
you know, explore
as a way to improve
these meta-heuristics,
particularly ant colony optimization.
So to present the method briefly,
the population is a population
inspired by the behavior of ants.
Each, you know,
agent would be
exploring one possible,
you know,
path
of solutions.
so it mimics how ants behave
in that
once
an ant or agent
finds
a promising path,
they're going to leave
a trail of

pheromones
that is going to be
more and more attractive
to other agents
as, you know,
as it is explored
by more ants.
Okay,
so the pheromone system
is a way for agents
to leave information
for others
and progressively
reinforce,
you know,
the best solutions.
And so
one subtlety
is that
there is also
a phenomenon
of evaporation
of these pheromones
as would happen naturally,
in fact,
so that
we don't get stuck
in local minima
and that,
you know,
we forget
bad solutions
over time.
And the choice
of the next node
to explore
in
the application

to the traveling
seismic problem,
for example,
would be based
on what
trail of pheromone
other ants left
and also
the intrinsic potential
of that
choice,
you know,
in finding
a minimum
length
traveling route.

Okay.

And so
this is the basic
ant colony
optimization,
but here again,
as you can see,
agents
are considered
to be passive,
you know,
receptors,
be it of pheromone
or information
in general.

They don't have
any intent
in their exploration.

And so
ant colony optimization
was a very successful
method for

the traveling seismic problem
up until very recently.
Now there are better
approximate methods.

And
as I was
briefly explaining,
the way
the ants
are going to
collectively
construct
a good route
would be that
each ant
visits
node by node
until it completes
one tour.

And
the most promising
tours are going
to have,
you know,
the most promising
edges,
in fact,
between nodes
are going to have
a higher
concentration of pheromone
over time.

And
at the end,
we keep the best
encountered solution
by the population
as well.

Now,
on the other hand,
active inference,
and here I speak
under your
supervision,
Daniel,
it's a shift
in the way
we view
perception
itself
from
just
reacting
to data
that is assumed
to be received
passively
to
considering
that
agents
have
an intent,
which is
to
predict
some outcome
and also
minimize
their
surprise
when
encountering
real outcomes,
the surprise
compared to

what they have
predicted.

So
there could
be
interesting
discussions
to be had
about
the statistical
models
that you
use
for
modeling
this prediction
and also
the surprise.

But
speaking again
abstractly
and at a
high level,
he will
assume that
agents anticipate
some future
state,
they build
a belief
about the
current
observations
they are
making,
and the
choices
they make

in the future
are not
just based
on the
immediate
data they
receive,
but also
on this
belief
that they
have
built
and
overall
that their
objective
would be
to minimize
free energy.
and so
the way
we implemented
this in
ant-colony
optimization
is that
it is going
to change
the way
we calculate
the node
selection
probability
which was
initially
just based
on

million
feromones
and here
we are going
to balance
the
feromone
level
available
in that
choice
with the
total
distance
of the
route
built
so far
and also
the current
belief
of the
agent
the ant
that is
exploring it
we also
have a
calculation
of free
energy
the typical
way
with
the logs
of B
in this
case

and we
will adjust
the belief
or confidence
level of
the ant
in its
current
exploration
part
based
on
the
length
of the
current
path
compared
to the
best
path
encountered
so far
by the
whole
population
and then
the
the
feromone
would be
updated
based
on
the
typical
way
based

on
the
length
of
the
current
path
and so
we're
going to
integrate
this
in
the
typical
scheme
for
ant colony
optimization
with these
changes
in the
way we
update
the way
we calculate
the probability
of selecting
the next
node
so the
typical
steps would
be that
we start
with some
uniform
paramount levels

for all
choices
the
ants
each one
of them
is going
to
update
its
belief
based
on
its
current
exploration
we calculate
the free
energy
for each
agent
and then
we're
going to
iteratively
update
both
the
beliefs
and
the
best
encountered
solutions
and
at the
end
we keep

the
best
route
discovered
by the
population
here again
there will be
some adjustments
to be made
about
the number
of
agents
in the
population
that you
would like
to consider
the amount
of computation
time that you
give yourself
that could
more or less
influence
the quality
of the
resulting
solutions
but again
abstractly
this is how
we would
structure
the algorithm
and so
we have

applied this
to the
traveling
salesman
problem
the
problem
I have
presented
here
let me
clear the
map
to take
another
example
and we
compare
this
enhanced
antcolony
optimization
as we call
it
or active
antcolony
optimization
with
both
basic
antcolony
optimization
and
the
nearest
neighbor
the very
naive

heuristic
as
baseline
so we
took
random
symmetric
graphs
of
increasing
sizes
and
one
interesting
thing
we
found
a few
interesting
things
it's that
first
this
simple
change
in the
way
you
calculate
the
probability
of
selecting
the
next
node
does
improve

the
quality
of
the
results
significantly
so it
does
give you
shorter
routes
on average
than
basic
antcolony
optimization
I don't
know if we
can see it
over this
example
let's try
so I
will run
the
basic
antcolony
optimization
here
we find
a route
of this
length
11286
with active
inference
seems to
be the

same
so maybe
we should
add more
nodes
and here
again
the
difference
is really
only appears
when you have
large enough
graphs
and complex
enough
graphs
let's try
again with
this one
maybe it
would be
luckier
I'll give
a comment
on that
while you're
running the
different
ones
yeah
and this
gives
possibly an
interesting
case
so first
off

there's
many
graphs
where a
number of
heuristics
could and
would and
should
converge on
the same
simple
output
so there's
never anything
wrong with
running multiple
parallel
meta-heuristics
and so I
was seeing
that free
energy
calculation
as
counteracting
with an
entropic
force
the convergent
tendency
of the
attractive
pheromone
so you have
the ants
laying down
pheromones

based upon
shorter
paths
and prior
success
and then
there's
this
kind
of
expansive
force
that's
keeping
an
additional
variance
element
around
and so
it's
kind
of
like
the
ball
is
rolling
to
the
bottom
of
the
hill
that's
the
standard
ant

colony
optimization
and then
the free
energy
is
keeping
it
a little
bit
off
the
bottom
so
then
looking
at
huge
numbers
of
replicates
for
graphs
it was
interesting
that
on
many
cases
it would
do
a tiny
bit
worse
but
then
in
several

cases
keeping
it
lifted
off
the
ground
with
that
entropic
force
would
actually
let it
explore
a totally
different
neighborhood
and find
vastly
better
solutions
so
that
overall
it
found
shorter
paths
but
that
it
found
basically
either
the
same
or

almost
the
same
in
most
cases
and
then
a few
situations
where
it
found
something
that
was
hugely
shorter
well
well
I have
two
comments
on this
that
was
actually
going
to
be
my
conclusion
for
this
presentation
the way
I see
it

is that
at this
point
it is
interesting
and useful
to do
applied
work
like we
do
here
to
show
in
practice
that
this
simple
shift
in
the
way
you
define
the
behavior
of
your
agents
does
improve
things
and
here
we
have
an

improvement

actually

with

this

graph

so

it

is

actually

significant

basic

ant

colony

gives

us

something

like

17

or

and

we

get

16

something

with

active

inference

so

there

is

definitely

value

in

applied

work

like

this

and

what
I would
be
interested
in
exploring
would
be
whether
this
also
replicates
for
other
population
based
meta-heuristics
notably
genetic
algorithms
or
other
approaches
simulated
and
healing
etc
but
the
point
you
raise
well
it's
little
beyond
my
understanding

of
active
inference
to be
honest
but
I
think
there
is
definitely
need
for
more
theoretical
grounding
and
it
sounds
very
interesting
it's
that
active
inference
just
this
change
in
the
way
you
can
create
probability
can
allow
you

to
escape
minima
that
are
not
that
permising
so
that
would
be
definitely
needed
at
some
point
of course
when
and if
we
get
promising
results
with
other
methods
is to
justify
it
theoretically
because
right
now
it's
just
based
on

intuition
indeed
agents
can be
improved
by having
this
intent
in their
exploration
but
I
would
be
very
interested
in
reading
more
about
the
relationship
with
entropy
and
how
it
can
be
seen
in
the
graph
of
the
tree
of
possible

solutions
that's
one
point
and
the
second
one
is
that
indeed
the
complexity
of the
graph
matters
of course
but
even
in
the
paper
and
let alone
in
my
toy
example
here
with
this
mini
app
we
are
still
considering
relatively

small
graphs
the
largest
graph
we
have
considered
was
500
nodes
which
remains
significant
you see
important
improvements
but
of course
the
real value
would come
only with
hundreds of
thousands
of nodes
that's when
you really
need
meta-heuristics
so
perhaps
I didn't
give
I wanted
to share
this as
well

you know
an overview
of
why this
problem
matters
why
the
traveling
salesman
problem
matters
first
theoretically
it
matters
because
as I
said
it
is
as
hard
as
any
problem
you
can
think
of
computationally
and
all
NP-hard
problems
reduce
to each
other

so
you
could
reduce
you
could
express
any
NP-hard
problem
as
a
traveling
salesman
and
also
because
it
has
well
practical
applications
it
could
be
literally
determining
routes
for trucks
or for
drones
but
more
it
is
being
used
in

chip
design
in
electronic
chips
design
that's
when
you
may
have
a
very
large
number
of
nodes
and
you
need
methods
that are
progressively
better
at the
limit
for
attending
towards
infinity
so
Daniel
you
have
an
idea
about
the

theoretical
grounding
of
the
method
well
I
mean
just
learning
from
you
about
this
broader
meta
heuristic
space
and
seeing
how
this
builds
on
prior
work
in
ants
optimization
and
in
active
and
for
ants
there
was
just

a few
interesting
pieces
and
I
agree
that
there
could
be
theoretical
and
empirical
exploration
to
flesh
out
some
of the
decisions
that we
made
here
like
the
bigger
picture
on
these
collective
intelligence
algorithms
and the
reason why
there's so
many
species
and so

many
different
metaphors
is because
there's
no
free
lunch
if
the
collective
behavioral
algorithm
recapitulates
important
features
of the
environment
then
it
might
be
able
to
do
well
however
if
it
doesn't
recapitulate
features
of the
environment
then
it
won't
necessarily

do
well
so
I'm
thinking
now
about
how
each
ant's
nest mate
knows
the best
solution
several
slides
I think
11
they
know
their
own
path
length
and
the
best
path
length
so
that's
an
example
of
an
abstraction
or a
calculation

that a
real
ant
nest
mate
wouldn't
necessarily
know
however
it might
have
expectations
about
path
length
from
its
own
learned
experience
or
from
evolution
so
I
think
that
this
design
space
of
how
do
we
recognize
natural
history
and

understand
what
the
cognitive
constraints
and what
the
cognitive
functions
are
in
nature
for
harvester
ants
and
turtle
ants
and
leaf
cutter
ants
just
going
to
nature
understanding
what
kinds
of
problems
are
being
solved
and
then
being
able

to
work
and
abstract
that
to
find
slightly
more
generic
forms
of those
collective
intelligence
methods
that
can
be
transferred
to
other
analogous
situations
like
more
generally
collecting
resources
or
information
or
finding
short
paths
and
then
what
other

elements
need
to
come
into
play
and
what
can
situationally
supercharge
the
algorithms
again
like
having
each
nestmate
know
at
all
times
whether
it's
itself
is
on a
short
or
long
path
yet
paths
of
real
ant
colonies
don't

have
that
feature
so
I
just
see
it
as
kind
of
like
a
mechanic
shop
with
all
these
different
options
and
meta
heuristics
how do
we
design
that
biological
metaphor
so
that
it
captures
relevant
features
of our
task
and

domain
because
there
could be
some
incredibly
efficient
algorithms
which
isn't
even to
say
they
would
be
super
transfer
but
there
might
be
some
really
specific
algorithms
that
could
have
incredible
interpretability
or performance
attributes
if we
can select
appropriately
among
collective
behavioral

algorithms
what could
really help
here
to feed
this
reflection
about
what could
be the
underlying
abstract
principles
that explain
this improvement
would be
perhaps if
we could
visualize
the tree
of
solutions
being
explored
for
small
scale
problems
like we
have
here
up to
a few
hundred
nodes
I think
it would
be very

interesting
to have
perhaps
some
colorful
visualization
of
basic
ACO
basic
optimization
versus
the active
version
and I would
really like
to explore
this idea
you have
mentioned
which is
that
this
small
change
you make
to the
way
the
probability
of
selecting
the
next
node
is
calculated
does

allow
you
to
escape
potentially
not
very
good
neighborhoods
so that's
definitely
something
we'll keep
in mind
in any
case
for
this
paper
and
the
so far
limited
experiments
we have
run
one
very
consistent
trend
is that
the
larger
the
graph
the
higher
the

percentage
of
improvements
so
you
only
see
the
real
value
for
large
scale
graphs
there
is
a
little
hiccup
here
for
100
node
for
some
reason
and
I
think
we
had
100
graphs
per
node
size
so
it

may
be
because
of
the
samples
or
in
any
case
we
can
see
the
trend
which
is
that
the
larger
the
size
of
the
graph
the
higher
the
percentage
of
improvement
but
on
the
other
hand
and
very

interestingly
the
computation
time
does not
increase
proportionally
to the
size
of the
graph
in fact
it
decreases
so
again
it's
not
much
adding
new
operations
to
the
method
new
steps
but
simply
changing
the way
one
operation
is
done
which
is
the

probability
of
selecting
the
next
node
so
obviously
that's
going
to
add
some
complexity
but
very
interestingly
this
decreases
with
the
size
of
the
graph
so
it
may
even
become
neglectable
for
very
large
scale
graphs
where
these

kind
of
improvements
matter
so
yes
please
yeah
I think
another
design
pattern
or something
to explore
is
having
more
simpler
nest
mates
versus
having
fewer
more
cognitively
sophisticated
nest
mates
and
that's
something
again
that
real
colonies
trade
off
in

the
wild
too
but
now
we can
explore
ranging
from
truly
multi-threadable
simple
nest
mates
no
information
sharing
not even
finding out
these
colony
level
summary
statistics
super
multi-thread
super
paralyzable
all the
way on
through
having
more
advanced
nest
mates
at the
per

thread
level
but maybe
they can
send each
other
messages
or they
can plan
ahead
explicitly
three
time
steps
but if
they're
planning
ahead
explicitly
three
time
steps
and
each
of
them
is
doing
a
little
tree
roll
out
at
each
time
step
then

the
computational
complexity
increases
a lot
and so
that
space
of
being
able
to
do
sweeps
across
more
simpler
to
fewer
more
complicated
nest
mates
or
cuckoos
or
whatever
the
metaphor
is
and then
understand
where are
we getting
the most
bang for
the buck
for a

given
type
of graph
maybe
if it's
a lattice
like graph
simpler
colonies
do better
but if
it's
a small
world
graph
topology
or
preferential
attachment
then
having
the
planning
capacity
so that
you can
know
that you're
going to
get out
of a
small
neighborhood
maybe
that
comes
into
play

again
so
it
just
shows
that
like
there
isn't
going
to be
a
general
approach
other
than
this
sort
of
meta
heuristic
design
level
of
generality
helping
us
articulate
which
collective
behavioral
methods
and how
they should
be
tuned
to
bring

to
a
given
problem
so
I
think
there
are
two
points
here
that
could
be
interesting
definitely
graph
topology
matters
that's
something
we definitely
need to
explore
whether
the
structure
of the
graph
you know
influences
the quality
of the
results
you get
and I
think

for
your
first
point
which
is
basically
what
is
the
most
bare
bones
population
search
method
we
could
explore
to
really
just
pinpoint
what
is
the
improvement
brought
by
active
inference
something
we could
also
see
would
be
I

think
genetic
algorithms
evolutionary
genetic
algorithms
evolution
is
really
the
most
general
metaphor
for
behavior
of
populations
and
there
you
don't
have
information
sharing
across
generations
you
just
have
message
transmission
in
the
genes
you
may
also
have

some
randomness
introduced
by
random
mutations
that can
happen
and that
can
help
you
escape
not
the
informizing
neighborhoods
and I
think
you wrote
a paper
with
if I'm
not
mistaken
about
an
application
of
active
inference
to
evolution
Daniel
right
yes
the
variational

synthesis
for
eco
evo
devo
exactly
and I
found a
very
interesting
idea
there
again
without
getting into
your
technicalities
it's that
what can
explain
the
evolution
of
species
or
populations
could also
explain
the
evolution
of
individuals
so
I
thought
it's
a
very

powerful
idea
that
you
have
this
multi-level
insight
and that's
perhaps
something
we could
also
explore
within
the
scope
of
genetic
algorithms
which are
really just
a simple
model
for evolution
that does
perform
very well
for
problems
like
the
TSP
yeah
just a
short
comment
on that

so
in
cognitive
sciences
the
unifying
imperative
is kind
of like
well why
should we
have a
separate
theory
of
anticipation
or memory
or attention
why not
seek this
unifying
approach
to model
various
cognitive
phenomena
so now
applying that
to
ecology
or to
biology
why
would we
have
one
method
or

analytical
framework
for
studying
dynamics
and change
within a
lineage
vertical
transmission
and then
have a
separate
theory
for
change
that happens
across
lineages
or
communication
across
lineages
like in
ecology
so
by
unifying
analogy
how could
we look
at
ecosystems
as
consisting
of these
composed
interacting

components
that are
changing
and
constituting
of
sub
components
as well
how can
we see
all of
those
processes
sometimes
which look
more like
vertical
lineage
based
transmission
other
times
which look
more like
horizontal
ecological
message
passing
and start
to look
at those
overall
dynamics
as part
of a
unified
at least

measurement

observation

modeling

approach

like

let's

just say

we were

to do

two

different

colonies

or two

populations

of colonies

and then

we have

them doing

different

algorithms

and then

we start

to let

that whole

thing play

out

we could

be talking

about the

variational

free energy

of the

nest

mate

within

the

colony

the

colony
as a
collective
with its
best
solution
the
populations
and
so on
will
that
yield
better
explanations
predictions
transferable
methods
that's
the question
but
that's
the
setup
it would
be very
interesting
to find
this
I think
that would
be the
most
at least
aesthetically
and
in terms
of

Occam's
Razor
simplest
theory
it's to
have
something
that
does
apply
indeed
for
both
the
individual
agents
as well
as
the
whole
population
and
here
another
in terms
of
elegance
and
beauty
another
thing
that's
also
helpful
is to
have
agents
share

as
little
information
as
possible
between
them
and
also
as
you
mentioned
before
plan
had
as
little
as
possible
which
also
has
of
course
practical
computational
implications
you
would
consume
less
computation
so
I
think
that
would
be

useful
for
that
to
have
this
unifying
theory
that
will
apply
both
for
the
behavior
of
individual
agents
as
well
as
the
population
as
a
whole
so
perhaps
something
to
explore
with
genetic
algorithms
!
we
can
demonstrate

it
even
more
potently
on
larger
scale
graphs
that
active
inference
does
improve
the
performance
of
recreation
based
meta
heuristics
with a
very
simple
change
again
that
is
very
cheap
computationally
that
may
even
disappear
become
neglectable
for
very

large
scale
graphs
this
additional
computational
cost
and
for
future
research
directions
I
think
we
have
mentioned
the
main
ones
it's
that
in
my
opinion
the
most
important
one
and
if
there's
someone
to
provide
this
I
think

it
would
be
yourself
Daniel
we
need
better
theoretical
grounding
for
what
is
going
on
we
could
definitely
of
course
inform
that
with
visualizations
like
I
mentioned
of
the
tree
of
solutions
and
how
it
is
explored
in

both
methods
but
it
understand
how
this
enhancement
occurs
fundamentally
in
abstract
terms
we
also
need
to
apply
to
other
meta
heuristics
I
think
the
best
candidates
as I
mentioned
would
be
first
genetic
algorithms
but
also
particle
swarm

optimization
which
has
its
own
peculiarities
we
also
need
simpler
and
more
bare bones
methods
this
is
just
an
engineered
belief
update
mechanism
but
there
is
certainly
a
simpler
and
more
elegant
way
to
state
the
equation
and
to go

even
beyond
population
based
search
and
I
know
that
many
people
are
working
on
this
in
the
active
inference
discord
what
about
other
multi-agent
systems
right
not
just
large
language
model
systems
but
in
general
multi-agent
systems
for

robotics
etc
could we
also
improve
their
collective
behavior
by
including
active
inference
that's
something
I would
also
love
to
explore
and
so
some
references
if you
are
interested
in
taking
it
further
first
our
paper
which
is
on
archive
and

also
on
preprints.
IU
but
the
advantage
on
preprints.
is
that
you
can
listen
to
podcasts
in
layman's
terms
about
the
paper
so
enhancing
is
here
you
can
download
the
AI
generated
podcast
to
get
layman's
overview
of the

paper
we
have
also
open
sourced
the
whole
code
and
process
from
the
beginning
in
both
our
githubs
so
active
aco
is
the
name
of
the
repo
and
we
have
this
mini
app
active
aco
at
dot
replete

dot
app
where
you
can
experiment
with
different
solutions
different
parts
on
real
maps
for
example
South
America
here
both
for
neighborhood
traveling
salesman
problems
as well
as
global
optimization
problems
and
compare
the
outcomes
of
active
aco
with

the
baseline
nearest
neighbor
so
this
is it
for
me
thank you
very much
for
the invitation
Daniel
and
the
listeners
if
any
for
their
attention
if
you
would
like
to
chat
about
this
paper
or
the
integration
of
active
inference
in

general
into
operations
research
problems
or
theoretical
CS
here
is
my
contact
thank
you
awesome
I'll
make a
comment
and then
read a
question
from the
chat
well
I also
really
like
all
these
accessory
fractal
all
these
accessory
pieces
to the
work
the

preprints
and also
the
replit
app
I
think
it
also
is
like
kind
of
building
out
this
meta
publication
that's
something
I learned
from you
as well
Daniel
was that
the PDF
is just
the beginning
right
of the
life cycle
of a
research
work
which is
absolutely
true
we tend

to see
it
as you
know
the
end
game
you
got
published
you
got
the
PDF
posted
somewhere
but
that's
really
just
the
beginning
of
you
know
and
the
real
fruits
may
may
only
appear
three
five
years
later
with

the
context
you
make
the
discussions
the
products
you
build
on top
of
it
so
that's
definitely
important
yeah
I
think
that's
super
cool
and
and
we
can
continue
making
that
really
multimedia
and
accessible
and
rigorous
so
it's

it's
like
on
that
sort
of
DCI
level
that's
really
exciting
okay
I'll
ask
a
question
from
the
chat
okay
Burt
wrote
would
this
approach
be
suited
for
multivariable
optimization
like
adding
in
travel
time
or
time
slots

so
multi
objective
right
yeah
what
we're
talking
about
and
here
I
think
well
that's
that's
pretty
much
my
PhD
topic
multi
objective
optimization
so
I
feel
very
strongly
about
the
kind
of
methods
that
can
work
there

meta
heuristics
in
general
I
mean
there
are
different
classes
of
meta
heuristics
that
work
better
for
multi
objective
optimization
but
this
is
definitely
something
to
explore
you
know
integrating
active
inference
notably
to
generate
the
as
you

know
multi
objective
optimization
problems
don't
have
one
optimal
solution
but
even
the
meaning
of
optimal
is
kind
of
mood
for
objective
problems
say
for example
that you
have a
car
that is
iterated
on both
its
price
its cost
and its
performance
well
what would

you do
if one
is good
the other
is not
that great
you would
only consider
efficient
solutions
right
so the
main
problem
is to
generate
or explore
this front
this large
potentially large
set of
solutions
that are not
dominated
and
I
wouldn't
make
any
guess
about
whether
active
inference
would
work
there
or

whether
this
method
would
be
translatable
but
as
a
research
product
it
could
be
really
a
PhD
or
postdoc
project
to
explore
the
use
of
active
inference
in
generating
and
exploring
the
part of
front
of
multi
objective
optimization

problems
so
perhaps
the
active
inference
institute
could
get
something
like
that
going
cool
yeah
what
comes
to
mind
there
one
would
be
keeping
the
kernel
method
the
same
and
using
program
logic
to
implement
other
constraints
like

adding
some
day
night
cycle
so
not even
trying
to
make
that
part
into
active
inference
but
just
add
some
secondary
logic
or
you
know
one
one
in
every
hundred
nodes
just
randomly
disappears
or
something
like
that
and

then
see
how
these
meta
heuristics
in a
more
changing
constraint
setting
adapt
to
changes
in the
environment
and then
the
more
fully
active
inference
approach
would
be
develop
more
sophisticated
generative
models
so
more
multi
variable
joint
distributions
for
nest

and
colony
perception
cognition
action
impact
and
then
explore
okay
as we
bring in
more
joint
variables
into the
distribution
obviously
it's taking
more computational
resources
it's a
higher
dimensional
front
to
explore
so
when
and how
should
we
pursue
that
full
joint
distribution
and

then
what
ways
would
there
be
factorize
it
down
to
solve
that
as
a
subset
of
linked
problems
and
how
do
we
find
which
subsetting
and
factorizing
of
linked
problems
allows
the
simplest
possible
nestmate
to
still
have

relevant
performance
and
then
we
sketch
out
those
subspaces
with
parameter
sweeps
and
sampling
the
reason
why
I
think
active
inference
is
honestly
even
more
promising
than
for
single
objective
problems
like
this
in
multi
objective
it's
that

another
dimension
if you
have
a car
for
example
that
costs
a lot
but
is
performance
another
one
that
is
cheap
but
not
that
great
is
that
there
is
no
objectively
optimal
solution
and
there
is
a
whole
class
of
methods

known
as
interactive
methods
where
you
try
to
integrate
the
preferences
of
decision
maker
into
the
exploration
method
so
it
is
really
a
person
guiding
the
exploration
with
their
own
preferences
and
here
we
are
touching
on
something

that
neuroscience
and
active
inference
is
familiar
with
which
is
how
to
integrate
reasoning
and
perception
so
I
would
be
very
interested
in
exploring
this
reading
about
it
if
anyone
is
inspired
to
write
about
it
how
to

integrate
active
inference
in
these
interactive
methods
that
are
typically
used
for
exploring
large-scale
parental prompts
yeah
one

one
I think
footnote
there
for the
the
curious
would
be
in
the
language
model
space
we have
these
reward
reinforcement
driven
models

that
then get
tuned
through
human
feedback
rlhf
with
humans
giving
a
rating
or a
preference
score
so you
have
the
algorithm
is
focusing
on
maximizing
satisfaction
and then
the
human
feedback
has to
do
with
do
I
like
this
more
or
less

we
could
be
exploring
some
interactive
optimization
methods
with more
expressivity
like
I'm
interested
in
paths
for the
traveling
salesman
that have
small
turning
angles
and then
that would
adjust
the
decision
making
if there
was a
turning
angle
variable
now
we're
focusing
on making
that

small
and then
we say
actually
I care
20%
about the
small
turning
angle
and 80%
about short
path length
overall
and so
be able
to have
this
interactive
blending
and a
more
expressive
epistemic
toolkit
for having
interactive
optimization
co-optimization
and that
could be
really
cool
or you know
a path
that is
just
aesthetically

pleasing
there are
actually
routing
methods
used in
tourism
that try
to
integrate
both
the
length
of the
route
but if
you are
designing
tourists
routes
you also
want it
to be
pleasant
so there
are
definitely
methods
that
are
applied
this
way
and
here
it's
a remark
in general

about
single
objective
optimization
it's
always
an
oversimplification
where you
say
that
this
is
the
one
objective
that
matters
it's
never
fully
the case
and there
are always
other dimensions
that can
be
integrated
now
are
percentages
or
just a
weighted
sum
of the
objectives
are

they
enough
they
are
more
often
than
not
limiting
because
there's
a whole
part
of the
part of
that you
will be
able to
reach
with
a weighted
sum
so it's
definitely
interesting
to
you mentioned
language models
you mentioned
reinforcement learning
from human
feedback
so these are
definitely
interesting
approaches
to integrate
in exploring

this
in general
having
multi-objective
optimization
2.0
with all
these
recent
advances
I think
is
sorely
needed
cool
okay
do you
have any
closing
comments
or
thoughts
or
well
my
last
thoughts
is that
it was
a very
pleasant
time
I thank
you very
much
for this
invitation
and

honestly
I thought
it was
just
five
minutes
or
ten
minutes
of
presentation
but
I look
at my
clock
and I
see
that
we
have
spent
almost
an hour
so it
is really
a rare
pleasure
to have
this
kind
of
experiences
where
you don't
feel
time
anymore
thank

you
very
much
Daniel
thank
you
awesome
okay
looking
forward
to
the
next
chapters
until
next
time
bye
!